

Unit 3: Boolean Algebra & Logic Gates

3.1 Boolean Algebra

Boolean algebra is a branch of mathematics that deals with binary variables (0 and 1) and logical operations. Basic operations are:

- AND ($\cdot$)
- OR ($+$)
- NOT ($\neg$)

Laws of Boolean Algebra:

1. Identity Law: $A + 0 = A$, $A \cdot 1 = A$
2. Null Law: $A + 1 = 1$, $A \cdot 0 = 0$
3. Idempotent Law: $A + A = A$, $A \cdot A = A$
4. Inverse Law: $A + A' = 1$, $A \cdot A' = 0$
5. Commutative Law: $A + B = B + A$, $A \cdot B = B \cdot A$
6. Associative Law: $(A + B) + C = A + (B + C)$, $(A \cdot B) \cdot C = A \cdot (B \cdot C)$
7. Distributive Law: $A \cdot (B + C) = A \cdot B + A \cdot C$, $A + (B \cdot C) = (A + B)(A + C)$
8. Absorption Law: $A + (A \cdot B) = A$, $A \cdot (A + B) = A$
9. De Morgan's Theorems: $(A \cdot B)' = A' + B'$, $(A + B)' = A' \cdot B'$

3.2 Logic Gates

Logic gates are electronic devices that perform basic logical functions. They form the building blocks of digital circuits.

- Basic Gates: AND, OR, NOT
- Universal Gates: NAND, NOR
- Exclusive Gates: XOR, XNOR

Each gate can be represented with a symbol, Boolean expression, and truth table.

3.3 Simplification of Boolean Expressions

Boolean expressions can be simplified using Boolean algebra laws and Karnaugh Maps (K-Map).

- 2-variable K-Map
- 3-variable K-Map
- 4-variable K-Map

Standard forms: SOP (Sum of Products) and POS (Product of Sums).

3.4 Important Questions

1. State and prove De Morgan's theorems.
2. Simplify the Boolean expression using K-map: $F(A,B,C,D) = \Sigma(0,1,2,5,7,8,9,10,14)$.
3. Explain XOR gate with truth table and applications.
4. What are universal gates? Why are they called universal?
5. Draw the circuit diagram for $(A + B) \cdot C'$ using logic gates.